

ZHEXIN(TONY) WU

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EDUCATION

Department of Mathematics, ETH Zürich	Doctoral Student, Sep 2026 – Present
Research: Optimal Execution, Market Microstructure & Reinforcement Learning	
Supervisor: Prof. Patrick Cheridito	
Management, Technology and Economics, ETH Zürich	Master of Science, Sep 2023 – Sep 2026
Focus: Quantitative Finance	GPA: 5.89/6.00
Information Technology and Electrical Engineering, ETH Zürich	Master of Science with Distinction, Sep 2021 – Jul 2023
Major: Signal Processing and Machine Learning	GPA: 5.76/6.00
Electrical and Computer Engineering, University of Michigan – Ann Arbor	Master of Science Program, Sep 2020 – May 2021
Major: Signal and Image Processing and Machine Learning (SIPML)	GPA: 4.00/4.00
Shen Yuan Honors College, Beihang University	Bachelor of Engineering, Sep 2015 – Jul 2019
Major: Electronic and Information Engineering	GPA: 3.80/4.00, Rank: 15/209

CERTIFICATIONS

CFA Program	Passed Level II, Jan 2026
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INTERNSHIPS

IPO Internship	Analyst
Sinolink Securities[code]	Jun 2024 to Sep 2024
- Developed scraper programs to collect and analyze the latest data on IPO statuses, prospectuses and inquiry letters from public websites - Developed a custom GPT to construct structured industry and company analysis reports	
Quantitative Internship in Factor Investment	Quantitative Analyst
China Securities[code]	Feb 2024 to Mar 2024
- Constructed CSI 300 and CSI 500 factor database using Tushare data terminal - Developed trading strategy with information-coefficient-weighted factor using Zipline framework	

RESEARCH EXPERIENCE

Multi-Agent Reinforcement Learning for Market Microstructure and Optimal Execution	
With Dr. Jean-Loup Dupret and Prof. Patrick Cheridito[paper]	Feb 2025 to Present
- Developed a novel multi-agent reinforcement learning (MARL) framework based on the Kyle model to study strategic interactions among heterogeneous groups of market participants and market efficiency through limit order book (LOB) dynamics - Extended the Almgren-Chriss model with RL-based agents and an endogenous price process for optimal execution - Refactored ABIDES-Gym to enable multi-agent training and asynchronous execution in a simulated LOB market	
In-Vivo Fetal Thalamus Parcellation	Project Collaborator & Software Developer
With Dr. Hui Ji and Prof. András Jakab[paper][code]	Mar 2023 to Dec 2023
- Unsupervised fetal thalamus functional group recognition with clustering algorithms - Feature engineering using coordinate and spherical harmonic coefficient from diffusion MRI - Optimized for high reproducibility between scan and re-scan data with image registration and Bayesian optimization - Implemented scree plot, silhouette score and gap statistic for choosing optimal number of clusters	
MR Image Reconstruction with Diffusion Models and Implicit Neural Representations	Master's Thesis
Advised by Dr. Valery Vishnevskiy and Prof. Sebastian Kozerke [paper][slides]code (diffusion models)[code (implicit NR)]	Oct 2022 to Jun 2023
- Developed score-based diffusion models for 2D and 2D + time CMR image reconstruction with high acceleration rates - Proposed proximal mapping for complex-valued image reconstruction algorithm with score-based diffusion models - Conducted extensive ablation study on type of algorithm (sampling or MAP), choice of prior (model-based or data-driven) and patching-artefacts removal strategies - Developed implicit neural representations for 2D, 2D + time and higher dimensional CMR image reconstruction	
Sequence Models in 3D MRI Brain Metastases Detection	Semester Thesis
Advised by Prof. Ender Konukoglu [paper][slides][code]	Apr 2022 to Jul 2022
- Task: Brain metastases detection in T_1-weighted 3D MR images - Revealed statistical significance of sequence models by training simple continuous patch-sequence classifier - Reduced false positives by incorporating active learning - Investigated the use of deep reinforcement learning agent	

Effects of Data Augmentation and Semi-Supervised Learning in Domain Generalization

Semester Thesis

Advised by Prof. Ender Konukoglu [paper][slides][code]

Oct 2021 to Mar 2022

- Task: 2D MRI cardiac image segmentation under domain-shift
- Investigated using consistency loss as both auxiliary loss for training and proxy loss for test-time adaptation
- Proposed a new inversion-based data augmentation
- Conducted rich experiments on different data augmentation schemes
- Investigated the use of meta-learning

Manifold Optimization & Neural Collapse

Research Assistant

Advised by Prof. Qing Qu[paper][code]

Jan 2021 to Jun 2021

- Created a smooth introduction to Riemannian submanifold optimization, with geometric illustration and concrete algorithmic examples.
- Participated in project A Geometric Analysis of Neural Collapse with Unconstrained Features and proof-read the script.
- Carried out experiments on neural collapse of CNN's last layer with oblique-manifold constraints on features and/or layer weights.

PROJECTS

Walmart Sales Prediction

Course Project

[executive summary][slides][code]

Mar 2024 to May 2024

- Developed machine learning pipeline following CRISP-DM, with an emphasis on explanatory model analysis
- Created executive recommendation from model prediction and variable importance

Model Fusion for Medical Image Segmentation via Optimal Transport

Course Project

[paper][code]

Sep 2022 to Jan 2023

- Extension of model fusion via optimal transport from cascade-like architecture (VGG, ResNet &etc) to more general architecture whose computational graph has richer adjacency structure
- Proposed model fusion approaches for prevalent U-Net and Transformer architectures used for medical image segmentation
- Comprehensive ablation study on fusion weighting scheme for typical medical data scenarios including distributed learning and domain-shift issue

Permutation-Invariant Variational Network for 2D + Time Cardiac Image Reconstruction

Course Project

[paper][code]

May 2022 to Jun 2022

- Implemented permutation-invariant variational network in PyTorch
- Implemented 2D and 3D variational network to make comparison with permutation-invariant variational network
- Re-organized derivation of optimization steps for loop-unrolling

Monitoring Social Distancing and Mask Wearing

EECS 504 Course Project

[paper][video][code]

Nov 2020 to Dec 2020

- Selected the topic and created the framework of the pipeline
- Responsible for human instance segmentation and face detection; proposed a chin locator based on the results of instance segmentation via Mask R-CNN
- Augmented face dataset with Gaussian Pyramid and integrated multiple datasets
- Improved the performance of mask classifier by designing a upsampler based on DCGAN with a group member
- Created prototype of social distance circle mapping using homography with a group member

Michigan Tourist Guide

SI 507 Course Project

[report][video][code]

Mar 2021 to Apr 2021

- Gathered data from multiple sources with web APIs and web crawling with BeautifulSoup4.
- Created a full-stack web app with Flask with rich interactive functionalities.

The 13th Mathematical Modeling Contest of Beijing Normal University

Team Leader, Second Prize

Advised By Assoc. Prof. Yingzhe Wang

Apr 2017 to May 2017

Project: Tourist-flow Prediction at South Luogu Lane in Beijing, China

- Constructed cellular automata model to simulate the changes in tourist-flow with MATLAB
- Later improved theoretical basis of the paper for the contest by constructing a model based on PDEs (Partial Differential Equations) with varied boundary conditions (compared to the original map); the polished work as the final course project achieved 1/209 of the whole class

SELECTED ONLINE COURSES AND PROJECTS

A full list of online courses & projects can be viewed here

- **Sparse Representations in Signal and Image Processing (by Technion)** Jul 2021 to Aug 2021
 - o **Topics:** Theoretical analysis of P_0 problem and guarantee of pursuit algorithms' stability; greedy pursuit algorithms and L_1 relaxation; FISTA and K-SVD algorithms(Dictionary Learning); application of sparseland model in image denoising, morphological components analysis(MCA), super resolution & etc.
 - o **Projects in Python:** OMP & Basis Pursuit for solving P_0 ; Image Reconstruction with OMP; Unitary Dictionary Learning; Image Denoising with Dictionary Learning; Image Deblurring

AWARDS

- National Scholarship for 2016-2017 academic year
- Merit Student for 2015-2016, 2016-2017 and 2017-2018 academic year
- First Prize of Learning Merit Scholarship for 2015-2016 and 2016-2017 academic year
- University Excellent Student for 2016-2017 academic year
- Second prize of The 13th Mathematical Modeling Contest of Beijing Normal University
- Third Prize of Academic Competition Scholarship
- First Prize of Excellent Social Work
- The 27th and 28th National High School Student Chemistry Competition, Second Prizes

SKILLS SUMMARY

Programming Languages Python, C/C++, Julia, MATLAB, Bash Script, Java, Python

Quant Frameworks Backtrader, VNPY, Zipline, Qlib

Machine Learning PyTorch, TensorFlow, JAX, RLlib, OpenCV

Statistical Analysis R, Stata, SPSS, Excel VBA

Financial Data Terminals LSEG Workspace, Wind Financial Terminal

Database Management SQL, PySpark

Web Development HTML5, CSS3, JavaScript, Bootstrap4, Django, Flask, Ruby on Rails

Hobbies Classical Piano, English Literature

Languages Chinese (Native), English (Fluent), Deutsch (Anfänger)

GRADUATE COURSES AT ETH ZÜRICH

401-6624-11L Applied Time Series

363-0392-00L Strategic Management

363-1077-00L Entrepreneurship

401-3929-00L Financial Risk Management in Social and Pension Insurance

363-0421-00L Management of Digital Transformation

363-0389-00L Technology and Innovation Management

401-3629-00L Quantitative Risk Management

363-0584-00L International Monetary Economics

363-1080-00L Responsible Leadership

401-3915-73L Machine Learning in Finance and Insurance[code]

401-4623-00L Time Series Analysis

363-0585-00L Intermediate Econometrics

363-1021-00L Monetary Policy

363-0560-00L Financial Management

363-0570-00L Principles of Econometrics

363-1098-00L Business Analytics[code]

363-0515-00L Decisions, Markets, and Games

363-0575-00L Economic Growth, Cycles and Policy

363-1063-00L Academic Writing Course

363-0711-00L Accounting for Managers

363-0503-00L Principles of Microeconomics

363-0565-00L Principles of Macroeconomics

363-0541-00L Economic Dynamics and Complexity

363-1004-00L Operations Research

363-0453-00L Strategic Supply Chain Management

363-0341-00L Introduction to Management

363-0403-00L Introduction to Marketing

263-3210-00L Deep Learning[code]

263-5210-00L Probabilistic Artificial Intelligence[code]

401-4944-20L Mathematics of Data Science

851-0252-15L Network Analysis

252-0220-00L Introduction to Machine Learning [code]

227-0391-00L Medical Image Analysis

227-0424-00L Model- and Learning-Based Inverse Problems in Imaging[code]

227-0948-00L Magnetic Resonance Imaging in Medicine

227-0449-00L Seminar in Biomedical Image Computing

401-3901-00L Linear & Combinatorial Optimization

401-3621-00L Fundamentals of Mathematical Statistics

227-0447-00L Image Analysis and Computer Vision

MOB-001 Basic German 1; A1: Intensive Course / Sprachenzentrum der UZH und der ETH Zürich

GRADUATE COURSES AT UMICH

EECS 501 Probability and Random Processes

EECS 559 Optimization Methods in Signal Processing and Machine Learning

SI 507 Intermediate Programming[code]

EECS 551 Matrix Methods for Signal Processing, Data Analysis and Machine Learning

EECS 504 Foundations of Computer Vision[code]

AUDITED COURSES

401-3904-22L Convex Optimization

EECS 598-005 Deep Learning for Computer Vision[code]